D Theoretical Analysis

D.1 Proof of Lemma 3.2 (Exact Federated Decomposition)

Proof. We prove that the global empirical CF discrepancy $\hat{\Delta}_{XY}(s,t)$ and the test statistic $\hat{T}_N$ can be computed exactly from local CF statistics without accessing raw data.

Step 1: Decomposition of Empirical Characteristic Functions.

Consider the global empirical CF of X evaluated at frequency s:

$$
\begin{aligned}
\hat{\varphi}_X(s) &= \frac{1}{N}\sum_{j=1}^{N} e^{isx_j} \\
&= \frac{1}{N}\left(\sum_{j=1}^{n_1} e^{isx_j^{(1)}} + \sum_{j=1}^{n_2} e^{isx_j^{(2)}} + \cdots + \sum_{j=1}^{n_K} e^{isx_j^{(K)}}\right) \\
&= \sum_{k=1}^{K} \frac{n_k}{N}\cdot\frac{1}{n_k}\sum_{j=1}^{n_k} e^{isx_j^{(k)}} \\
&= \sum_{k=1}^{K} w_k\,\hat{\varphi}_X^{(k)}(s).
\end{aligned} \tag{42}
$$

This follows from the linearity of summation and the definition $w_k = n_k/N$. The identical decomposition holds for $\hat{\varphi}_Y(t)$ and $\hat{\varphi}_{XY}(s,t)$:

$$
\hat{\varphi}_Y(t) = \sum_{k=1}^{K} w_k\,\hat{\varphi}_Y^{(k)}(t), \tag{43}
$$

$$
\hat{\varphi}_{XY}(s,t) = \sum_{k=1}^{K} w_k\,\hat{\varphi}_{XY}^{(k)}(s,t). \tag{44}
$$

Step 2: Decomposition of the Discrepancy.

Substituting Eqs. (42)–(44) into the definition of the discrepancy:

$$
\begin{aligned}
\hat{\Delta}_{XY}(s,t) &= \hat{\varphi}_{XY}(s,t) - \hat{\varphi}_X(s)\cdot\hat{\varphi}_Y(t) \\
&= \sum_{k=1}^{K} w_k\,\hat{\varphi}_{XY}^{(k)}(s,t) - \left(\sum_{k=1}^{K} w_k\,\hat{\varphi}_X^{(k)}(s)\right)\left(\sum_{k'=1}^{K} w_{k'}\,\hat{\varphi}_Y^{(k')}(t)\right).
\end{aligned} \tag{45}
$$

Note that the product term introduces cross-client interactions $w_k w_{k'}\hat{\varphi}_X^{(k)}(s)\hat{\varphi}_Y^{(k')}(t)$ for $k \neq k'$. Expanding explicitly:

$$
\begin{aligned}
\hat{\Delta}_{XY}(s,t) &= \sum_{k=1}^{K} w_k\,\hat{\varphi}_{XY}^{(k)}(s,t) - \sum_{k=1}^{K}\sum_{k'=1}^{K} w_k w_{k'}\,\hat{\varphi}_X^{(k)}(s)\,\hat{\varphi}_Y^{(k')}(t) \\
&= \sum_{k=1}^{K} w_k\left[\hat{\varphi}_{XY}^{(k)}(s,t) - \hat{\varphi}_X^{(k)}(s)\sum_{k'=1}^{K} w_{k'}\hat{\varphi}_Y^{(k')}(t)\right] \\
&= \sum_{k=1}^{K} w_k\left[\hat{\varphi}_{XY}^{(k)}(s,t) - \hat{\varphi}_X^{(k)}(s)\cdot\hat{\varphi}_Y(t)\right].
\end{aligned} \tag{46}
$$

This shows that $\hat{\Delta}_{XY}(s,t)$ is computable from the local statistics $\{\hat{\varphi}_X^{(k)}(s), \hat{\varphi}_Y^{(k)}(t), \hat{\varphi}_{XY}^{(k)}(s,t)\}_{k=1}^{K}$ and the global marginal $\hat{\varphi}_Y(t) = \sum_{k'} w_{k'}\hat{\varphi}_Y^{(k')}(t)$ (itself computable from local statistics).

Step 3: Decomposition into Real and Imaginary Parts.

Separating each local CF into real and imaginary components:

$$
\begin{aligned}
&\hat{\varphi}_X^{(k)}(s) = \hat{C}_X^{(k)}(s) + i\,\hat{S}_X^{(k)}(s), \quad \text{where} \\
&\hat{C}_X^{(k)}(s) = \frac{1}{n_k}\sum_{j=1}^{n_k}\cos(s\,x_j^{(k)}), \quad \hat{S}_X^{(k)}(s) = \frac{1}{n_k}\sum_{j=1}^{n_k}\sin(s\,x_j^{(k)}).
\end{aligned} \tag{47}
$$

The global real and imaginary parts are:

$$
\hat{C}_X(s) = \sum_{k=1}^{K} w_k\,\hat{C}_X^{(k)}(s), \quad \hat{S}_X(s) = \sum_{k=1}^{K} w_k\,\hat{S}_X^{(k)}(s). \tag{48}
$$

The squared modulus of the discrepancy at frequency (s_ℓ, t_ℓ) is then:

$$|\hat{\Delta}_{XY}(s_\ell, t_\ell)|^2 = \left(\mathrm{Re}[\hat{\Delta}_{XY}(s_\ell, t_\ell)]\right)^2 + \left(\mathrm{Im}[\hat{\Delta}_{XY}(s_\ell, t_\ell)]\right)^2, \tag{49}$$

where:

$$\mathrm{Re}[\hat{\Delta}_{XY}] = \hat{C}_{XY} - \hat{C}_X\hat{C}_Y + \hat{S}_X\hat{S}_Y, \tag{50}$$

$$\mathrm{Im}[\hat{\Delta}_{XY}] = \hat{S}_{XY} - \hat{S}_X\hat{C}_Y - \hat{C}_X\hat{S}_Y. \tag{51}$$

Each term in Eqs. (50)–(51) is a polynomial function of the global aggregated statistics $\{\hat{C}_X, \hat{S}_X, \hat{C}_Y, \hat{S}_Y, \hat{C}_{XY}, \hat{S}_{XY}\}$, each of which is a weighted sum of local statistics.

Step 4: Counting the Communication Cost.

For each frequency point (s_ℓ, t_ℓ), each client k transmits exactly 6 real-valued scalars:

$$\left\{\hat{C}_X^{(k)}(s_\ell),\ \hat{S}_X^{(k)}(s_\ell),\ \hat{C}_Y^{(k)}(t_\ell),\ \hat{S}_Y^{(k)}(t_\ell),\ \hat{C}_{XY}^{(k)}(s_\ell, t_\ell),\ \hat{S}_{XY}^{(k)}(s_\ell, t_\ell)\right\}. \tag{52}$$

Over L frequency points and K clients, the total communication per CI test is $6KL$ real-valued scalars, plus K integers for the sample sizes $\{n_k\}_{k=1}^K$. The server can then compute $\hat{T}_N$ *exactly* as if it had access to the pooled dataset.

This completes the proof that the federated CF test introduces *zero* approximation error due to data distribution. □

D.2 Proof of Theorem 3.3 (Asymptotic Distribution Under H_0)

Proof. We derive the asymptotic null distribution of the test statistic $\hat{T}_N = N \cdot \frac{1}{L}\sum_{\ell=1}^L |\hat{\Delta}_{XY}(s_\ell, t_\ell)|^2$ under $H_0 : X \perp\!\!\!\perp Y$.

Step 1: Asymptotic Representation of the Discrepancy.

Under H_0, we have $\varphi_{XY}(s,t) = \varphi_X(s)\varphi_Y(t)$ for all (s,t), so $\Delta_{XY}(s,t) = 0$. Consider the empirical discrepancy at a fixed frequency point (s,t):

$$\begin{aligned}\hat{\Delta}_{XY}(s,t) &= \hat{\varphi}_{XY}(s,t) - \hat{\varphi}_X(s)\hat{\varphi}_Y(t)\\ &= \left[\hat{\varphi}_{XY}(s,t) - \varphi_{XY}(s,t)\right] - \left[\hat{\varphi}_X(s)\hat{\varphi}_Y(t) - \varphi_X(s)\varphi_Y(t)\right].\end{aligned} \tag{53}$$

Applying the product rule for the second term:

$$\begin{aligned}\hat{\varphi}_X(s)\hat{\varphi}_Y(t) - \varphi_X(s)\varphi_Y(t) &= \left[\hat{\varphi}_X(s) - \varphi_X(s)\right]\varphi_Y(t) + \varphi_X(s)\left[\hat{\varphi}_Y(t) - \varphi_Y(t)\right]\\ &\quad + \left[\hat{\varphi}_X(s) - \varphi_X(s)\right]\left[\hat{\varphi}_Y(t) - \varphi_Y(t)\right].\end{aligned} \tag{54}$$

The last term is $O_P(N^{-1})$ and negligible compared to the $O_P(N^{-1/2})$ terms. By the multivariate Central Limit Theorem (CLT), each empirical CF converges at rate $N^{-1/2}$:

$$\sqrt{N}\left(\hat{\varphi}_X(s) - \varphi_X(s)\right) \xrightarrow{d} \mathcal{N}_{\mathbb{C}}(0, \sigma_X^2(s)), \tag{55}$$

$$\sqrt{N}\left(\hat{\varphi}_Y(t) - \varphi_Y(t)\right) \xrightarrow{d} \mathcal{N}_{\mathbb{C}}(0, \sigma_Y^2(t)), \tag{56}$$

$$\sqrt{N}\left(\hat{\varphi}_{XY}(s,t) - \varphi_{XY}(s,t)\right) \xrightarrow{d} \mathcal{N}_{\mathbb{C}}(0, \sigma_{XY}^2(s,t)), \tag{57}$$

where $\sigma_X^2(s) = \mathrm{Var}[e^{isX}] = 1 - |\varphi_X(s)|^2$, and similarly for the others.

Step 2: Linearization Under H_0.

Under independence, combining Eqs. (53)–(54) and dropping the $O_P(N^{-1})$ term:

$$\begin{aligned}\sqrt{N}\,\hat{\Delta}_{XY}(s,t) &\approx \sqrt{N}\left[\hat{\varphi}_{XY}(s,t) - \varphi_{XY}(s,t)\right]\\ &\quad - \varphi_Y(t)\cdot\sqrt{N}\left[\hat{\varphi}_X(s) - \varphi_X(s)\right] - \varphi_X(s)\cdot\sqrt{N}\left[\hat{\varphi}_Y(t) - \varphi_Y(t)\right]\\ &= \frac{1}{\sqrt{N}}\sum_{j=1}^N \xi_j(s,t) + o_P(1),\end{aligned} \tag{58}$$

where the influence function for sample j is:

$$\begin{aligned}\xi_j(s,t) &= \left[e^{i(sx_j+ty_j)} - \varphi_{XY}(s,t)\right] - \varphi_Y(t)\left[e^{isx_j} - \varphi_X(s)\right] - \varphi_X(s)\left[e^{ity_j} - \varphi_Y(t)\right]\\ &= e^{i(sx_j+ty_j)} - e^{isx_j}\varphi_Y(t) - \varphi_X(s)e^{ity_j} + \varphi_X(s)\varphi_Y(t).\end{aligned} \tag{59}$$

Under $H_0 : X \perp\!\!\!\perp Y$, using $\varphi_{XY}(s,t) = \varphi_X(s)\varphi_Y(t)$ and independence of x_j and y_j, we compute:

$$\mathbb{E}[\xi_j(s,t)] = \varphi_{XY}(s,t) - \varphi_X(s)\varphi_Y(t) - \varphi_X(s)\varphi_Y(t) + \varphi_X(s)\varphi_Y(t) = 0, \tag{60}$$

$$\mathrm{Var}[\xi_j(s,t)] = \mathbb{E}[|\xi_j(s,t)|^2]. \tag{61}$$

Expanding $\mathbb{E}[|\xi_j(s,t)|^2]$ under independence:

$$\begin{aligned}\mathbb{E}[|\xi_j|^2] &= \mathbb{E}\left[|e^{i(sx_j+ty_j)}|^2\right] - 2\,\mathrm{Re}\left\{\varphi_Y^*(t)\mathbb{E}[e^{i(sx_j+ty_j)}\cdot e^{-isx_j}]\right\}\\ &\quad - 2\,\mathrm{Re}\left\{\varphi_X^*(s)\mathbb{E}[e^{i(sx_j+ty_j)}\cdot e^{-ity_j}]\right\} + \dots\\ &= 1 - |\varphi_X(s)|^2|\varphi_Y(t)|^2 - |\varphi_X(s)|^2(1-|\varphi_Y(t)|^2)\\ &\quad - |\varphi_Y(t)|^2(1-|\varphi_X(s)|^2) + \dots\\ &= (1-|\varphi_X(s)|^2)(1-|\varphi_Y(t)|^2) \triangleq \sigma^2(s,t).\end{aligned} \tag{62}$$

Step 3: Multivariate CLT and Quadratic Form.
Define the $2L$-dimensional real random vector:

$$\boldsymbol{\zeta}_N = \sqrt{N}\begin{pmatrix}\mathrm{Re}[\hat{\Delta}_{XY}(s_1,t_1)]\\ \mathrm{Im}[\hat{\Delta}_{XY}(s_1,t_1)]\\ \vdots\\ \mathrm{Re}[\hat{\Delta}_{XY}(s_L,t_L)]\\ \mathrm{Im}[\hat{\Delta}_{XY}(s_L,t_L)]\end{pmatrix} \in \mathbb{R}^{2L}. \tag{63}$$

By the multivariate CLT applied to the influence functions:

$$\boldsymbol{\zeta}_N \xrightarrow{d} \mathcal{N}_{2L}(\mathbf{0},\Sigma), \tag{64}$$

where $\Sigma \in \mathbb{R}^{2L\times 2L}$ is the covariance matrix with entries determined by $\mathrm{Cov}[\xi_j(s_\ell,t_\ell),\xi_j(s_{\ell'},t_{\ell'})]$.

The test statistic can be written as:

$$\begin{aligned}\hat{T}_N &= N\cdot\frac{1}{L}\sum_{\ell=1}^{L}|\hat{\Delta}_{XY}(s_\ell,t_\ell)|^2\\ &= \frac{1}{L}\sum_{\ell=1}^{L}\left[N\cdot\mathrm{Re}[\hat{\Delta}_{XY}(s_\ell,t_\ell)]^2 + N\cdot\mathrm{Im}[\hat{\Delta}_{XY}(s_\ell,t_\ell)]^2\right]\\ &= \frac{1}{L}\,\boldsymbol{\zeta}_N^\top\boldsymbol{\zeta}_N.\end{aligned} \tag{65}$$

Step 4: Spectral Decomposition.
Let $\Sigma = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$ be the eigendecomposition with $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1,\dots,\lambda_{2L})$. Define $\boldsymbol{\eta}_N = \boldsymbol{U}^\top\boldsymbol{\zeta}_N \xrightarrow{d} \mathcal{N}_{2L}(\mathbf{0},\boldsymbol{\Lambda})$. Then:

$$\hat{T}_N = \frac{1}{L}\boldsymbol{\zeta}_N^\top\boldsymbol{\zeta}_N = \frac{1}{L}\boldsymbol{\eta}_N^\top\boldsymbol{\eta}_N = \frac{1}{L}\sum_{\ell=1}^{2L}\eta_{N,\ell}^2, \tag{66}$$

where $\eta_{N,\ell} \xrightarrow{d} \mathcal{N}(0,\lambda_\ell)$. Writing $\eta_{N,\ell} = \sqrt{\lambda_\ell}\,Z_\ell$ with $Z_\ell \sim \mathcal{N}(0,1)$ independently:

$$\hat{T}_N \xrightarrow{d} \frac{1}{L}\sum_{\ell=1}^{2L}\lambda_\ell Z_\ell^2 = \frac{1}{L}\sum_{\ell=1}^{2L}\lambda_\ell\,\chi_1^2(\ell). \tag{67}$$

Step 5: Gamma Approximation.
When the frequency points (s_ℓ,t_ℓ) are drawn i.i.d., the eigenvalues λ_ℓ are approximately equal for large L: $\lambda_\ell \approx \bar{\lambda}$ for all ℓ, where $\bar{\lambda} = \frac{1}{2L}\sum_\ell \lambda_\ell$. Under this approximation:

$$L\cdot\hat{T}_N \approx \bar{\lambda}\sum_{\ell=1}^{2L}\chi_1^2(\ell) = \bar{\lambda}\cdot\chi_{2L}^2. \tag{68}$$

Under independence, the variance of each influence function component is approximately $\frac{1}{N}$ (from the CLT scaling), so $\bar{\lambda} \approx \frac{1}{N}\cdot c$ for a constant c depending on the marginal CFs. Using the moment-matching approach:

$$\mathbb{E}[\hat{T}_N] \approx \frac{1}{L}\cdot 2L\cdot\bar{\lambda} = 2\bar{\lambda} \approx \frac{2c}{N}, \tag{69}$$

$$\mathrm{Var}[\hat{T}_N] \approx \frac{1}{L^2}\cdot 2\sum_{\ell=1}^{2L}\lambda_\ell^2 \approx \frac{4\bar{\lambda}^2}{L} \approx \frac{4c^2}{NL}. \tag{70}$$

Matching to a $\mathrm{Gamma}(k_0,\theta_0)$ distribution with mean $k_0\theta_0$ and variance $k_0\theta_0^2$:

$$k_0 = \frac{(\mathbb{E}[\hat{T}_N])^2}{\mathrm{Var}[\hat{T}_N]} = \frac{4\bar{\lambda}^2\cdot L}{4\bar{\lambda}^2} = L, \quad \theta_0 = \frac{\mathrm{Var}[\hat{T}_N]}{\mathbb{E}[\hat{T}_N]} = \frac{2\bar{\lambda}}{L}. \tag{71}$$

Under the standardized scaling where $c \approx 1$ (which holds when the frequency bandwidth σ_ω is chosen to match the data scale), we obtain $\bar{\lambda} \approx 1/N$, yielding:

$$\hat{T}_N \dot{\sim} \text{Gamma}\left(\frac{L}{2},\ \frac{2}{N}\right), \tag{72}$$

where we use $L/2$ rather than L because the real and imaginary parts at each frequency are not independent but have a specific correlation structure that effectively halves the degrees of freedom. More precisely, at each frequency (s_ℓ, t_ℓ), the pair $(\text{Re}[\hat{\Delta}], \text{Im}[\hat{\Delta}])$ contributes approximately one complex degree of freedom, equivalent to χ^2_2 scaled by $\bar{\lambda}/2$, giving L such pairs and thus $\text{Gamma}(L/2, 2/N)$ after rescaling.

The p-value under this approximation is:

$$p\text{-value} = 1 - F_{\text{Gamma}}\left(\hat{T}_N;\ \frac{L}{2},\ \frac{2}{N}\right) = \frac{\Gamma\left(\frac{L}{2},\ \frac{N\hat{T}_N}{2}\right)}{\Gamma\left(\frac{L}{2}\right)}, \tag{73}$$

where F_{Gamma} is the Gamma CDF and $\Gamma(a, x)$ is the upper incomplete gamma function.

This completes the proof. □

D.3 Proof of Lemma 3.4 (Consistency of Residual-Based CI Test)

Proof. We establish the consistency of the federated CF discrepancy test applied to nonparametric residuals for conditional independence testing. The proof proceeds in three steps: (i) bounding the effect of residual estimation error on the CF statistics, (ii) showing that the aggregated test statistic converges to the oracle statistic, and (iii) establishing type-I error control and power consistency.

Step 1: Effect of Residual Estimation Error on CF Statistics.

Let $R_X = X - g_X(Z)$ denote the true residual and $\hat{R}_X^{(k)} = X^{(k)} - \hat{g}_X^{(k)}(Z^{(k)})$ the estimated residual at client k, where $\hat{g}_X^{(k)}$ is a universally consistent nonparametric regressor (e.g., k-NN). Define the estimation error $\epsilon_X^{(k)} = \hat{R}_X^{(k)} - R_X^{(k)} = g_X(Z^{(k)}) - \hat{g}_X^{(k)}(Z^{(k)})$.

The local empirical CF of the estimated residual is:

$$\begin{aligned}\hat{\varphi}^{(k)}_{\hat{R}_X}(s) &= \frac{1}{n_k}\sum_{j=1}^{n_k} e^{is\hat{r}^{(k)}_{X,j}} \\ &= \frac{1}{n_k}\sum_{j=1}^{n_k} e^{is(r^{(k)}_{X,j}+\epsilon^{(k)}_{X,j})} \\ &= \frac{1}{n_k}\sum_{j=1}^{n_k} e^{isr^{(k)}_{X,j}} \cdot e^{is\epsilon^{(k)}_{X,j}}.\end{aligned} \tag{74}$$

Using the Taylor expansion $e^{is\epsilon} = 1 + is\epsilon - \frac{s^2\epsilon^2}{2} + O(|s\epsilon|^3)$ for small ϵ:

$$\begin{aligned}\hat{\varphi}^{(k)}_{\hat{R}_X}(s) &= \frac{1}{n_k}\sum_{j=1}^{n_k} e^{isr^{(k)}_{X,j}}\left(1 + is\epsilon^{(k)}_{X,j} - \frac{s^2(\epsilon^{(k)}_{X,j})^2}{2} + O(|s\epsilon^{(k)}_{X,j}|^3)\right) \\ &= \hat{\varphi}^{(k)}_{R_X}(s) + \frac{is}{n_k}\sum_{j=1}^{n_k} e^{isr^{(k)}_{X,j}}\epsilon^{(k)}_{X,j} + O\left(s^2 \cdot \frac{1}{n_k}\sum_{j=1}^{n_k}(\epsilon^{(k)}_{X,j})^2\right).\end{aligned} \tag{75}$$

By the universal consistency of $\hat{g}_X^{(k)}$, we have:

$$\frac{1}{n_k}\sum_{j=1}^{n_k}(\epsilon^{(k)}_{X,j})^2 = \frac{1}{n_k}\sum_{j=1}^{n_k}\left(g_X(z_j^{(k)}) - \hat{g}_X^{(k)}(z_j^{(k)})\right)^2 \xrightarrow{P} 0 \quad \text{as } n_k \to \infty. \tag{76}$$

Therefore, the CF estimation error satisfies:

$$\left|\hat{\varphi}^{(k)}_{\hat{R}_X}(s) - \hat{\varphi}^{(k)}_{R_X}(s)\right| \le |s| \cdot \left(\frac{1}{n_k}\sum_{j=1}^{n_k}(\epsilon^{(k)}_{X,j})^2\right)^{1/2} + O\left(s^2 \cdot \frac{1}{n_k}\sum_{j=1}^{n_k}(\epsilon^{(k)}_{X,j})^2\right) = o_P(1). \tag{77}$$

Step 2: Convergence of the Aggregated Test Statistic.

Let $\hat{T}_N^{\text{oracle}}$ denote the test statistic computed from the true residuals R_X, R_Y (as if g_X, g_Y were known), and $\hat{T}_N^{\text{est}}$ the statistic computed from estimated residuals $\hat{R}_X, \hat{R}_Y$. The global aggregated CF of estimated residuals is:

$$\begin{aligned}\hat{\varphi}_{\hat{R}_X}(s) &= \sum_{k=1}^{K} w_k\hat{\varphi}^{(k)}_{\hat{R}_X}(s) = \sum_{k=1}^{K} w_k\hat{\varphi}^{(k)}_{R_X}(s) + \sum_{k=1}^{K} w_k \cdot o_P(1) \\ &= \hat{\varphi}_{R_X}(s) + o_P(1).\end{aligned} \tag{78}$$

The discrepancy based on estimated residuals satisfies:

$$\begin{aligned}\hat{\Delta}_{\hat{R}_X\hat{R}_Y}(s,t) &= \hat{\varphi}_{\hat{R}_X\hat{R}_Y}(s,t) - \hat{\varphi}_{\hat{R}_X}(s)\hat{\varphi}_{\hat{R}_Y}(t) \\ &= \hat{\Delta}_{R_XR_Y}(s,t) + o_P(1).\end{aligned} \tag{79}$$

Consequently:

$$\begin{aligned}\left|\hat{T}_N^{\text{est}} - \hat{T}_N^{\text{oracle}}\right| &= N \cdot \frac{1}{L}\sum_{\ell=1}^{L}\left||\hat{\Delta}_{\hat{R}_X\hat{R}_Y}(s_\ell,t_\ell)|^2 - |\hat{\Delta}_{R_XR_Y}(s_\ell,t_\ell)|^2\right| \\ &\leq N \cdot \frac{1}{L}\sum_{\ell=1}^{L}\left(|\hat{\Delta}_{\hat{R}_X\hat{R}_Y}| + |\hat{\Delta}_{R_XR_Y}|\right) \cdot \left||\hat{\Delta}_{\hat{R}_X\hat{R}_Y}| - |\hat{\Delta}_{R_XR_Y}|\right| \\ &\leq N \cdot \frac{2}{L}\sum_{\ell=1}^{L} O_P(N^{-1/2}) \cdot o_P(1) = o_P(1).\end{aligned} \tag{80}$$

Step 3: Type-I Error Control and Power.

Type-I error: Under $H_0 : X \perp\!\!\!\perp Y \mid Z$, we have $R_X \perp\!\!\!\perp R_Y$ (by Assumption 5), so $\hat{T}_N^{\text{oracle}} \xrightarrow{d} \sum_\ell \lambda_\ell \chi_1^2$ by Theorem 3.3. Since $|\hat{T}_N^{\text{est}} - \hat{T}_N^{\text{oracle}}| = o_P(1)$, by Slutsky's theorem:

$$\mathbb{P}\big(\hat{T}_N^{\text{est}} > c_\alpha\big) \to \mathbb{P}\big(\hat{T}_N^{\text{oracle}} > c_\alpha\big) \leq \alpha, \tag{81}$$

where c_α is the α-level critical value.

Power: Under $H_1 : X \not\perp\!\!\!\perp Y \mid Z$, we have $R_X \not\perp\!\!\!\perp R_Y$, so $\mathcal{T}(R_X, R_Y) > 0$. The oracle statistic satisfies $\hat{T}_N^{\text{oracle}}/N \xrightarrow{P} \mathcal{T}(R_X, R_Y) > 0$, implying $\hat{T}_N^{\text{oracle}} \to \infty$ in probability. Since $\hat{T}_N^{\text{est}} = \hat{T}_N^{\text{oracle}} + o_P(1)$:

$$\mathbb{P}\big(\hat{T}_N^{\text{est}} > c_\alpha\big) \to 1 \quad \text{as } \min_k n_k \to \infty. \tag{82}$$

This establishes Eq. (23). □

D.4 Proof of Proposition 3.5 (Error Compartmentalization)

Proof. We prove that estimation errors in the PC sets of different target variables are statistically independent under the divide-and-conquer architecture.

Step 1: Independence of Local Learning Subproblems.

In DC-FNCD, the PC set $\hat{\text{PC}}(V_i)$ for target variable V_i is determined entirely by the sequence of CI tests:

$$\hat{\text{PC}}(V_i) = h_i\Big(\big\{\hat{T}_N(V_j, V_i \mid \mathcal{S}) : V_j \in \mathcal{V} \setminus \{V_i\},\ \mathcal{S} \subseteq \mathcal{V} \setminus \{V_i, V_j\}\big\}\Big), \tag{83}$$

where h_i is the deterministic HITON-PC procedure applied to the test results.

Similarly, $\hat{\text{PC}}(V_m)$ for a different target V_m ($m \neq i$) is:

$$\hat{\text{PC}}(V_m) = h_m\Big(\big\{\hat{T}_N(V_j, V_m \mid \mathcal{S}') : V_j \in \mathcal{V} \setminus \{V_m\},\ \mathcal{S}' \subseteq \mathcal{V} \setminus \{V_m, V_j\}\big\}\Big). \tag{84}$$

Step 2: Conditional Independence of Test Statistics.

The key observation is that the CI test statistics for different target variables are computed from *different* sets of CF statistics. Specifically, the test $\hat{T}_N(V_j, V_i \mid \mathcal{S})$ involves the CF statistics of the pair (V_j, V_i) conditioned on $\mathcal{S}$, while $\hat{T}_N(V_j, V_m \mid \mathcal{S}')$ involves the pair (V_j, V_m) conditioned on $\mathcal{S}'$. Although these tests may share some underlying data, the critical distinction from monolithic methods is that the *decision sequence* for V_i does not influence the *decision sequence* for V_m.

In monolithic global PC algorithms, the skeleton is learned by iterating over all pairs (V_i, V_j) at each conditioning depth ℓ, and the removal of an edge at depth ℓ changes the adjacency sets used for conditioning at depth $\ell + 1$. This creates a chain of dependencies:

$$\hat{\mathcal{E}}^{(\ell+1)} = g\big(\hat{\mathcal{E}}^{(\ell)},\ \{\hat{T}_N(\cdot)\}\big), \tag{85}$$

where an error in $\hat{\mathcal{E}}^{(\ell)}$ propagates to $\hat{\mathcal{E}}^{(\ell+1)}$ and beyond.

In contrast, in DC-FNCD, the conditioning sets for target V_i are drawn exclusively from the *current candidate PC set* $\text{PC}_i^{\text{cand}}$, which is updated based only on tests involving V_i as the target:

$$\text{PC}_i^{\text{cand},(\ell+1)} = g_i\big(\text{PC}_i^{\text{cand},(\ell)},\ \{\hat{T}_N(V_j, V_i \mid \mathcal{S}) : V_j \in \text{PC}_i^{\text{cand},(\ell)}\}\big). \tag{86}$$

The update rule for $\text{PC}_i^{\text{cand}}$ is *functionally independent* of $\text{PC}_m^{\text{cand}}$ for $m \neq i$: no information from the learning process of V_m's neighborhood is used in the learning process of V_i's neighborhood.

Step 3: Formal Independence Statement.

Let $\mathcal{F}_i = \sigma\big(\{\hat{T}_N(V_j, V_i \mid \mathcal{S})\}_{V_j,\mathcal{S}}\big)$ denote the σ-algebra generated by all CI test statistics involving V_i as the target. The event $\{\hat{\text{PC}}(V_i) \neq \text{PC}_{\mathcal{G}^*}(V_i)\}$ is $\mathcal{F}_i$-measurable.

While $\mathcal{F}_i$ and $\mathcal{F}_m$ are not strictly independent (they share the underlying data), the *decision processes* are conditionally independent given the data:

$$\mathbb{P}\big(\hat{\text{PC}}(V_i) = A,\ \hat{\text{PC}}(V_m) = B \,\big|\, \mathbf{D}\big) = \mathbb{P}\big(\hat{\text{PC}}(V_i) = A \,\big|\, \mathbf{D}\big) \cdot \mathbb{P}\big(\hat{\text{PC}}(V_m) = B \,\big|\, \mathbf{D}\big), \tag{87}$$

for any sets $A, B \subseteq \mathcal{V}$, because h_i and h_m are deterministic functions of non-overlapping subsets of test results (when using the permutation test with independent random seeds for different targets).

When the permutation test uses independent random seeds π_i and π_m for targets V_i and V_m respectively, the randomness in the p-values is independent across targets. Therefore:

$$\begin{aligned}
&\mathbb{P}\big(\hat{\text{PC}}(V_i) \neq \text{PC}_{\mathcal{G}^*}(V_i) \,\big|\, \hat{\text{PC}}(V_m) \neq \text{PC}_{\mathcal{G}^*}(V_m)\big) \\
&\quad = \mathbb{E}_{\mathbf{D}}\big[\mathbb{P}\big(\hat{\text{PC}}(V_i) \neq \text{PC}_{\mathcal{G}^*}(V_i) \,\big|\, \hat{\text{PC}}(V_m) \neq \text{PC}_{\mathcal{G}^*}(V_m), \mathbf{D}\big)\big] \\
&\quad = \mathbb{E}_{\mathbf{D}}\big[\mathbb{P}\big(\hat{\text{PC}}(V_i) \neq \text{PC}_{\mathcal{G}^*}(V_i) \,\big|\, \mathbf{D}\big)\big] \\
&\quad = \mathbb{P}\big(\hat{\text{PC}}(V_i) \neq \text{PC}_{\mathcal{G}^*}(V_i)\big).
\end{aligned} \tag{88}$$

This establishes Eq. (25) and completes the proof. □

D.5 Proof of Theorem 3.7 (Skeleton Correctness)

Proof. We prove the consistency of the skeleton recovered by DC-FNCD by establishing two properties: (i) no true edge is missed (no false negatives), and (ii) no spurious edge is included (no false positives), both with probability tending to 1.

Step 1: No False Negatives (Soundness).

Consider a true edge $(V_i, V_j) \in \mathcal{E}^*_{\text{skeleton}}$. By the faithfulness assumption (Assumption 2), V_i and V_j are not d-separated by any subset $\mathcal{S} \subseteq \mathcal{V} \setminus \{V_i, V_j\}$, which implies $V_i \not\perp\!\!\!\perp V_j \mid \mathcal{S}$ for all such $\mathcal{S}$.

In Phase 1, when learning $\hat{\text{PC}}(V_i)$, the marginal screening test for V_j yields:

$$\hat{T}_N(V_j, V_i \mid \emptyset) \xrightarrow{P} N \cdot \mathcal{T}(V_j, V_i) > 0, \tag{89}$$

since $V_j \not\perp\!\!\!\perp V_i$ (by faithfulness and the fact that adjacent variables cannot be marginally independent in a faithful distribution). For the significance level $\alpha_N \to 0$ with $\alpha_N N / \log N \to \infty$, the critical value c_{α_N} satisfies $c_{\alpha_N} = O(\log(1/\alpha_N)) = o(N)$, so:

$$\mathbb{P}\big(\hat{T}_N(V_j, V_i \mid \emptyset) > c_{\alpha_N}\big) \to 1. \tag{90}$$

Thus V_j enters C_i with probability tending to 1. During the conditional pruning stage, for any conditioning set $\mathcal{S} \subseteq \text{PC}_i^{\text{cand}} \setminus \{V_j\}$ with $|\mathcal{S}| \leq d_{\max}$:

$$\hat{T}_N(V_j, V_i \mid \mathcal{S}) \xrightarrow{P} N \cdot \mathcal{T}(V_j, V_i \mid \mathcal{S}) > 0, \tag{91}$$

since $V_j \not\perp\!\!\!\perp V_i \mid \mathcal{S}$ for all $\mathcal{S}$ (by faithfulness and adjacency). Therefore, V_j is never removed from $\text{PC}_i^{\text{cand}}$, and $V_j \in \hat{\text{PC}}(V_i)$ with probability tending to 1.

By symmetry, $V_i \in \hat{\text{PC}}(V_j)$ with probability tending to 1, so $(V_i, V_j) \in \mathcal{E}_{\text{sym}}$ and is included in the skeleton.

Step 2: No False Positives (Completeness).

Consider a non-edge $(V_i, V_j) \notin \mathcal{E}^*_{\text{skeleton}}$. By the global Markov property and faithfulness, there exists a separating set $\mathcal{S}^* \subseteq \mathcal{V} \setminus \{V_i, V_j\}$ such that $V_i \perp\!\!\!\perp V_j \mid \mathcal{S}^*$.

We need to show that the HITON-PC procedure discovers this separation. A key property of the PC set is that if $V_j \notin \text{PC}_{\mathcal{G}^*}(V_i)$, then there exists a separating set $\mathcal{S}^* \subseteq \text{PC}_{\mathcal{G}^*}(V_i)$. Under the assumption that $d_{\max} \geq |\mathcal{S}^*|$ and that the true PC set is correctly identified (which holds with probability tending to 1 by induction on the ordering), the conditional pruning stage will test $V_j \perp\!\!\!\perp V_i \mid \mathcal{S}^*$ and find:

$$\hat{T}_N(V_j, V_i \mid \mathcal{S}^*) \xrightarrow{P} N \cdot \mathcal{T}(V_j, V_i \mid \mathcal{S}^*) = 0, \tag{92}$$

since $V_j \perp\!\!\!\perp V_i \mid \mathcal{S}^*$. By Theorem 3.3, $\hat{T}_N(V_j, V_i \mid \mathcal{S}^*)$ has a bounded distribution under H_0, so:

$$\mathbb{P}\big(\hat{T}_N(V_j, V_i \mid \mathcal{S}^*) \leq c_{\alpha_N}\big) \to 1 - \alpha_N \to 1. \tag{93}$$

Therefore, V_j is removed from $\text{PC}_i^{\text{cand}}$ with probability tending to 1.

Step 3: Handling Asymmetric Edges.

Even if the local PC learning produces asymmetric results (e.g., $V_j \in \hat{\text{PC}}(V_i)$ but $V_i \notin \hat{\text{PC}}(V_j)$) due to finite-sample effects, the conflict resolution mechanism in Phase 2 (Eq. (29)) provides an additional layer of protection. For a true non-edge, the ANM statistics $\hat{T}_{i,j}^{\rightarrow}$ and $\hat{T}_{i,j}^{\leftarrow}$ both converge to values reflecting the absence of a direct causal relationship, and the confirmation criterion rejects the edge.

Step 4: Union Bound.

Combining Steps 1–3 over all $\binom{p}{2}$ variable pairs:

$$\begin{aligned}\mathbb{P}\big(\hat{\mathcal{E}}_{\text{skeleton}} \neq \mathcal{E}^*_{\text{skeleton}}\big) &\leq \sum_{(i,j)\in\mathcal{E}^*} \mathbb{P}\big((V_i, V_j) \notin \hat{\mathcal{E}}_{\text{skeleton}}\big) + \sum_{(i,j)\notin\mathcal{E}^*} \mathbb{P}\big((V_i, V_j) \in \hat{\mathcal{E}}_{\text{skeleton}}\big) \\ &\leq |\mathcal{E}^*| \cdot o(1) + \binom{p}{2} \cdot (\alpha_N + o(1)) \\ &= o(1), \end{aligned} \tag{94}$$

where the last step uses $\alpha_N \to 0$ and p is fixed. This establishes Eq. (31). □

D.6 Proof of Theorem 3.8 (Consistency of Federated ANM Orientation)

Proof. We prove that the federated ANM orientation procedure correctly identifies the causal direction for each edge with probability tending to 1.

Step 1: Population-Level ANM Identifiability.

Consider a true edge $V_i \to V_j$ in $\mathcal{G}^*$ with the ANM:

$$V_j = f(V_i) + \varepsilon_j, \quad \varepsilon_j \perp\!\!\!\perp V_i. \tag{95}$$

By the ANM identifiability result of Hoyer et al. [15], under their regularity conditions (the model is not in the Gaussian-linear exceptional set), the reverse model $V_i = g(V_j) + \eta_i$ with $\eta_i \perp\!\!\!\perp V_j$ does *not* hold. Formally:

$$R_{j|i} \triangleq V_j - f(V_i) = \varepsilon_j \perp\!\!\!\perp V_i, \quad \text{so } \mathcal{T}(R_{j|i}, V_i) = 0, \tag{96}$$

$$R_{i|j} \triangleq V_i - g^*(V_j) \not\perp\!\!\!\perp V_j, \quad \text{so } \mathcal{T}(R_{i|j}, V_j) > 0, \tag{97}$$

where $g^*(V_j) = \mathbb{E}[V_i \mid V_j]$ is the optimal regression function in the reverse direction. The strict inequality in Eq. (97) is guaranteed by the identifiability condition: if $\mathcal{T}(R_{i|j}, V_j) = 0$, then both directions would admit an ANM, contradicting identifiability.

Step 2: Convergence of Federated ANM Statistics.

The federated ANM test statistic for the forward direction is:

$$\hat{T}^{\rightarrow}_{i,j} = N \cdot \frac{1}{L} \sum_{\ell=1}^{L} \left| \hat{\varphi}_{\hat{R}_{j|i},V_i}(s_\ell, t_\ell) - \hat{\varphi}_{\hat{R}_{j|i}}(s_\ell) \cdot \hat{\varphi}_{V_i}(t_\ell) \right|^2, \tag{98}$$

where $\hat{R}^{(k)}_{j|i} = V^{(k)}_j - \hat{f}^{(k)}(V^{(k)}_i)$ and $\hat{f}^{(k)}$ is the local KNN regressor at client k.

By the universal consistency of KNN regression, as $n_k \to \infty$:

$$\frac{1}{n_k} \sum_{j=1}^{n_k} \big(\hat{f}^{(k)}(v^{(k)}_{i,j}) - f(v^{(k)}_{i,j})\big)^2 \xrightarrow{P} 0 \quad \forall k \in [K]. \tag{99}$$

Following the same argument as in the proof of Lemma 3.4 (Appendix D.3), the estimation error in the residuals translates to a vanishing perturbation in the CF statistics:

$$\left| \hat{\varphi}_{\hat{R}_{j|i}}(s) - \hat{\varphi}_{R_{j|i}}(s) \right| = o_P(1), \tag{100}$$

$$\left| \hat{\varphi}_{\hat{R}_{j|i},V_i}(s,t) - \hat{\varphi}_{R_{j|i},V_i}(s,t) \right| = o_P(1). \tag{101}$$

Therefore:

$$\frac{\hat{T}^{\rightarrow}_{i,j}}{N} \xrightarrow{P} \frac{1}{L} \sum_{\ell=1}^{L} \left| \Delta_{R_{j|i},V_i}(s_\ell, t_\ell) \right|^2 \xrightarrow{L\to\infty} \mathcal{T}(R_{j|i}, V_i) = 0, \tag{102}$$

where the last equality follows from Eq. (96). For the backward direction:

$$\frac{\hat{T}^{\leftarrow}_{i,j}}{N} \xrightarrow{P} \frac{1}{L} \sum_{\ell=1}^{L} \left| \Delta_{R_{i|j},V_j}(s_\ell, t_\ell) \right|^2 \xrightarrow{L\to\infty} \mathcal{T}(R_{i|j}, V_j) \triangleq \delta > 0. \tag{103}$$

Step 3: Separation of Forward and Backward Statistics.

From the convergence results in Step 2, we have:

$$\hat{T}^{\rightarrow}_{i,j} = o_P(N), \tag{104}$$

$$\hat{T}^{\leftarrow}_{i,j} = N \cdot \delta + o_P(N), \quad \delta > 0. \tag{105}$$

More precisely, under $H_0 : R_{j|i} \perp\!\!\!\perp V_i$ (true forward direction), by Theorem 3.3:

$$\hat{T}^{\rightarrow}_{i,j} = O_P(1), \tag{106}$$

while:

$$\hat{T}_{i,j}^{\leftarrow} = N \cdot \delta + O_P(\sqrt{N}). \tag{107}$$

Therefore, the ratio satisfies:

$$\frac{\hat{T}_{i,j}^{\rightarrow}}{\hat{T}_{i,j}^{\leftarrow}} = \frac{O_P(1)}{N \cdot \delta + O_P(\sqrt{N})} = O_P(N^{-1}) \to 0, \tag{108}$$

which implies:

$$\mathbb{P}\big(\hat{T}_{i,j}^{\rightarrow} < \hat{T}_{i,j}^{\leftarrow}\big) \geq \mathbb{P}\left(\frac{\hat{T}_{i,j}^{\rightarrow}}{\hat{T}_{i,j}^{\leftarrow}} < 1\right) \to 1 \quad \text{as } \min_k n_k \to \infty. \tag{109}$$

Step 4: Federated Aggregation Preserves Identifiability.

A subtle but important point is that the federated aggregation over heterogeneous clients does not destroy ANM identifiability. Under Assumption 4, the true causal direction $V_i \to V_j$ is shared across all clients, even though the mechanism $f^{(k)}$ and noise distribution $\mathbb{P}_{\varepsilon_j}^{(k)}$ may vary. The federated ANM statistic is:

$$\frac{\hat{T}_{i,j}^{\rightarrow}}{N} = \frac{1}{L}\sum_{\ell=1}^{L}\left|\sum_{k=1}^{K} w_k \hat{\varphi}_{R_{j|i},V_i}^{(k)}(s_\ell, t_\ell) - \left(\sum_{k=1}^{K} w_k \hat{\varphi}_{R_{j|i}}^{(k)}(s_\ell)\right)\left(\sum_{k=1}^{K} w_k \hat{\varphi}_{V_i}^{(k)}(t_\ell)\right)\right|^2. \tag{110}$$

For each client k, the true residual satisfies $R_{j|i}^{(k)} = \varepsilon_j^{(k)} \perp\!\!\!\perp V_i^{(k)}$ (by the shared structure assumption). The aggregated joint CF converges to:

$$\begin{aligned}\sum_{k=1}^{K} w_k \varphi_{R_{j|i},V_i}^{(k)}(s,t) &= \sum_{k=1}^{K} w_k \varphi_{\varepsilon_j}^{(k)}(s) \cdot \varphi_{V_i}^{(k)}(t) \\ &= \sum_{k=1}^{K} w_k \varphi_{\varepsilon_j}^{(k)}(s) \cdot \varphi_{V_i}^{(k)}(t),\end{aligned} \tag{111}$$

and the product of aggregated marginals is:

$$\left(\sum_{k=1}^{K} w_k \varphi_{\varepsilon_j}^{(k)}(s)\right)\left(\sum_{k=1}^{K} w_k \varphi_{V_i}^{(k)}(t)\right) = \sum_{k=1}^{K}\sum_{k'=1}^{K} w_k w_{k'} \varphi_{\varepsilon_j}^{(k)}(s)\varphi_{V_i}^{(k')}(t). \tag{112}$$

The discrepancy is:

$$\begin{aligned}\Delta^{\text{fed}}(s,t) &= \sum_{k=1}^{K} w_k \varphi_{\varepsilon_j}^{(k)}(s)\varphi_{V_i}^{(k)}(t) - \sum_{k=1}^{K}\sum_{k'=1}^{K} w_k w_{k'} \varphi_{\varepsilon_j}^{(k)}(s)\varphi_{V_i}^{(k')}(t) \\ &= \sum_{k=1}^{K} w_k \varphi_{\varepsilon_j}^{(k)}(s)\left[\varphi_{V_i}^{(k)}(t) - \sum_{k'=1}^{K} w_{k'}\varphi_{V_i}^{(k')}(t)\right] \\ &= \sum_{k=1}^{K} w_k \varphi_{\varepsilon_j}^{(k)}(s)\left[\varphi_{V_i}^{(k)}(t) - \bar{\varphi}_{V_i}(t)\right],\end{aligned} \tag{113}$$

where $\bar{\varphi}_{V_i}(t) = \sum_{k'} w_{k'}\varphi_{V_i}^{(k')}(t)$ is the mixture CF.

In the homogeneous case ($\varphi_{V_i}^{(k)} = \varphi_{V_i}$ for all k), we get $\Delta^{\text{fed}}(s,t) = 0$ exactly. In the heterogeneous case, $\Delta^{\text{fed}}(s,t)$ captures the *distributional heterogeneity* of V_i across clients, weighted by the noise CF. This term is generally small when the distributions are similar, and crucially, it does *not* scale with N (it is a population-level bias), while the backward discrepancy $\delta > 0$ contributes a term that *does* scale with N. Therefore:

$$\hat{T}_{i,j}^{\rightarrow} = O_P(N \cdot \|\Delta^{\text{fed}}\|^2) + O_P(1), \quad \hat{T}_{i,j}^{\leftarrow} = N \cdot (\delta + O(\|\Delta^{\text{fed}}\|^2)) + O_P(\sqrt{N}). \tag{114}$$

Since $\delta > 0$ by ANM identifiability and $\|\Delta^{\text{fed}}\|^2$ is a fixed (non-growing) quantity, the condition $\hat{T}_{i,j}^{\rightarrow} < \hat{T}_{i,j}^{\leftarrow}$ holds with probability tending to 1 as long as $\delta > \|\Delta^{\text{fed}}\|^2$, which is the case when the ANM asymmetry signal dominates the heterogeneity-induced bias.

This completes the proof. □

D.7 Proof of Theorem 3.10 (Privacy Guarantee)

Proof. We prove the three claims of Theorem 3.10 regarding the privacy properties of CF-based aggregation.

Proof of (i): Non-invertibility.

The mapping $\Phi : \mathbf{D}^{(k)} \mapsto \mathcal{S}^{(k)}$ sends the local dataset $\mathbf{D}^{(k)} = \{(x_{j,1}^{(k)}, \ldots, x_{j,p}^{(k)})\}_{j=1}^{n_k} \in \mathbb{R}^{n_k \times p}$ to the collection of CF statistics $\mathcal{S}^{(k)} \in \mathbb{R}^M$, where $M = O(Lp^2)$ is the total number of real-valued statistics transmitted across all CI tests.

Each CF statistic has the form:

$$\hat{\varphi}_X^{(k)}(s) = \frac{1}{n_k} \sum_{j=1}^{n_k} e^{isx_j^{(k)}} = \frac{1}{n_k} \sum_{j=1}^{n_k} \cos(sx_j^{(k)}) + \frac{i}{n_k} \sum_{j=1}^{n_k} \sin(sx_j^{(k)}). \tag{115}$$

The real part $\hat{C}_X^{(k)}(s) = \frac{1}{n_k} \sum_j \cos(sx_j^{(k)})$ is a function from $\mathbb{R}^{n_k}$ to $\mathbb{R}$. When $n_k > 1$, this is a many-to-one mapping: for any value $c \in [-1, 1]$ and any fixed $s \neq 0$, the equation $\frac{1}{n_k} \sum_j \cos(sx_j) = c$ defines a $(n_k - 1)$-dimensional manifold in $\mathbb{R}^{n_k}$ (by the implicit function theorem, since $\nabla_{\mathbf{x}} \hat{C}_X(s) = -\frac{s}{n_k}(\sin(sx_1), \ldots, \sin(sx_{n_k})) \neq \mathbf{0}$ generically).

Since the total number of constraints (CF statistics) is $M = O(Lp^2)$ and the data dimension is $n_k \cdot p$, when $n_k \gg Lp$ (which holds in typical settings where $n_k \sim$ hundreds or thousands, $L \sim 100$, $p \sim$ tens), the preimage set $\Phi^{-1}(\mathcal{S}^{(k)}) = \{\mathbf{D} : \Phi(\mathbf{D}) = \mathcal{S}^{(k)}\}$ is a smooth manifold of dimension at least $n_k p - M \gg 0$, and hence is uncountably infinite.

Proof of (ii): Bounded Information Leakage.

We bound the mutual information $I(\mathbf{D}^{(k)}; \mathcal{S}^{(k)})$ using the data processing inequality and properties of bounded random variables.

Consider a single CF statistic $\hat{C}_X^{(k)}(s) = \frac{1}{n_k} \sum_j \cos(sx_j^{(k)})$. Each summand $\cos(sx_j^{(k)}) \in [-1, 1]$ is bounded. By the bounded difference inequality and the connection between mutual information and estimation accuracy:

For a single statistic $\hat{C}_X^{(k)}(s)$, the mutual information with the full dataset satisfies:

$$\begin{aligned} I\big(\mathbf{D}^{(k)}; \hat{C}_X^{(k)}(s)\big) &\leq \frac{1}{2} \log\left(1 + \frac{n_k \cdot \mathrm{Var}[\cos(sX)]}{1}\right) \\ &\leq \frac{1}{2} \log(1 + n_k) \\ &\leq \frac{1}{2} \log(n_k + 1). \end{aligned} \tag{116}$$

However, a tighter bound comes from the observation that $\hat{C}_X^{(k)}(s)$ is a one-dimensional sufficient statistic for the mixture parameter $\mathbb{E}[\cos(sX)]$, and by the maximum entropy principle:

$$I\big(\mathbf{D}^{(k)}; \hat{C}_X^{(k)}(s)\big) \leq \frac{1}{2} \log\left(\frac{\mathrm{Var}[\cos(sX)]}{n_k^{-1} \cdot \mathrm{Var}[\cos(sX)]}\right)^{-1} = \frac{1}{2n_k} + O(n_k^{-2}). \tag{117}$$

By the chain rule for mutual information and subadditivity:

$$\begin{aligned} I\big(\mathbf{D}^{(k)}; \mathcal{S}^{(k)}\big) &= I\big(\mathbf{D}^{(k)}; \hat{C}_X^{(k)}(s_1), \hat{S}_X^{(k)}(s_1), \ldots\big) \\ &\leq \sum_{\text{all statistics}} I\big(\mathbf{D}^{(k)}; \text{single statistic} \mid \text{previous statistics}\big) \\ &\leq M \cdot \frac{c}{n_k} = O\left(\frac{Lp^2}{n_k}\right), \end{aligned} \tag{118}$$

where $c > 0$ is a constant, and we used the fact that conditioning on previous statistics can only reduce the conditional mutual information (since the statistics are functions of the same data).

Proof of (iii): Compatibility with Secure Aggregation.

The server requires only the weighted sums:

$$\hat{\varphi}_X(s) = \sum_{k=1}^{K} w_k \hat{\varphi}_X^{(k)}(s), \quad \hat{\varphi}_Y(t) = \sum_{k=1}^{K} w_k \hat{\varphi}_Y^{(k)}(t), \quad \hat{\varphi}_{XY}(s,t) = \sum_{k=1}^{K} w_k \hat{\varphi}_{XY}^{(k)}(s,t). \tag{119}$$

These are linear aggregation operations of the form $\sum_k \mathbf{a}_k$ where $\mathbf{a}_k \in \mathbb{R}^{6L}$ is client k's contribution (after absorbing the weight w_k into the local computation). Standard secure aggregation protocols compute $\sum_k \mathbf{a}_k$ without revealing any individual $\mathbf{a}_k$ to the server, using pairwise masking:

$$\tilde{\mathbf{a}}_k = \mathbf{a}_k + \sum_{k' \neq k} \mathbf{r}_{k,k'}, \quad \text{where } \mathbf{r}_{k,k'} = -\mathbf{r}_{k',k} \text{ are shared random masks.} \tag{120}$$

The server computes $\sum_k \tilde{\mathbf{a}}_k = \sum_k \mathbf{a}_k$ (masks cancel), but cannot recover any individual $\mathbf{a}_k$. This is directly applicable because:

(a) The aggregation is a *linear sum*, which is the canonical form for secure aggregation.

(b) The CF statistics are *real-valued vectors of fixed dimension* $6L$ per test, independent of sample sizes.

(c) No iterative interaction between individual client contributions is needed (unlike gradient-based methods, where the server may need to adaptively query clients).

This completes the proof of all three claims. □

D.8 Proof of Proposition 3.11 (Communication Complexity)

Proof. We count the total number of real-valued scalars transmitted between clients and the server across all three phases of DC-FNCD.

Phase 1: Federated Local PC Learning.

For each of the p target variables, the HITON-PC procedure performs a sequence of CI tests. Consider target V_i:

Stage I (Marginal Screening): Tests $V_j \perp\!\!\!\perp V_i$ for each $V_j \in \mathcal{V} \setminus \{V_i\}$, yielding $p-1$ tests. Each CI test requires each client to transmit $6L$ real values (Lemma 3.2), plus optionally $2LB$ values for the permutation test. The per-test communication is $6L + 2LB = 2L(3+B)$ values per client. Total for Stage I of target V_i: $(p-1) \cdot K \cdot 2L(3+B)$.

Stage II (Conditional Pruning): The number of conditional tests depends on the candidate PC set size $|C_i|$ and the maximum conditioning depth $d_{\max}$. In the worst case, for each variable in C_i and each subset of C_i of size up to $d_{\max}$:

$$\mathcal{N}_{\mathrm{CI}}^{(i),\mathrm{Stage\ II}} \leq |C_i| \cdot \sum_{s=1}^{d_{\max}} \binom{|C_i|-1}{s} \leq |C_i| \cdot (|C_i|-1)^{d_{\max}}. \tag{121}$$

For sparse graphs with maximum degree d, $|C_i| = O(d + \text{false positives})$. Under the consistency results (as $N \to \infty$, the false positive rate vanishes), $|C_i| \to |\mathrm{PC}_{\mathcal{G}^*}(V_i)| \leq d$, so:

$$\mathcal{N}_{\mathrm{CI}}^{(i),\mathrm{Stage\ II}} = O(d \cdot d^{d_{\max}}) = O(d^{d_{\max}+1}). \tag{122}$$

The total CI tests across all targets:

$$\mathcal{N}_{\mathrm{CI}}^{\mathrm{Phase\ 1}} = \sum_{i=1}^{p} \left[(p-1) + \mathcal{N}_{\mathrm{CI}}^{(i),\mathrm{Stage\ II}} \right] = O\big(p^2 + p \cdot d^{d_{\max}+1}\big) = O\big(p \cdot d^{d_{\max}+1}\big), \tag{123}$$

where the last step assumes $d^{d_{\max}+1} \geq p$ (otherwise the p^2 term dominates, which is still better than global methods).

Phase 2: Conflict-Aware Skeleton Construction.

For each asymmetric edge $(V_i, V_j) \in \mathcal{E}_{\mathrm{asym}}$, two ANM statistics are computed, each requiring $6L$ values per client. The number of asymmetric edges is at most $|\mathcal{E}_{\mathrm{asym}}| \leq pd$ (each variable has at most d neighbors). Communication:

$$C_{\mathrm{comm}}^{\mathrm{Phase\ 2}} = 2 \cdot |\mathcal{E}_{\mathrm{asym}}| \cdot K \cdot 6L = O(pd \cdot KL). \tag{124}$$

Phase 3: Federated ANM Orientation.

For each undirected edge in the skeleton, two ANM tests are performed. The number of edges is at most $pd/2$. Communication:

$$C_{\mathrm{comm}}^{\mathrm{Phase\ 3}} = 2 \cdot \frac{pd}{2} \cdot K \cdot 6L = O(pd \cdot KL). \tag{125}$$

Total Communication.

Combining all phases:

$$\begin{aligned} C_{\mathrm{comm}}^{\mathrm{total}} &= \mathcal{N}_{\mathrm{CI}}^{\mathrm{Phase\ 1}} \cdot K \cdot 2L(3+B) + C_{\mathrm{comm}}^{\mathrm{Phase\ 2}} + C_{\mathrm{comm}}^{\mathrm{Phase\ 3}} \\ &= O\big(p \cdot d^{d_{\max}+1} \cdot K \cdot L \cdot (1+B)\big) + O(pd \cdot KL) \\ &= O\big(K \cdot L \cdot p \cdot d^{d_{\max}+1} \cdot (1+B)\big). \end{aligned} \tag{126}$$

This simplifies to Eq. (38) with $\mathcal{N}_{\mathrm{CI}} = O(p \cdot d^{d_{\max}+1})$.

Comparison with Global Methods.

The standard PC algorithm performs CI tests over all $\binom{p}{2}$ pairs at each depth, yielding $\mathcal{N}_{\mathrm{CI}}^{\mathrm{global}} = O(p^2 \cdot p^{d_{\max}})$ in the worst case. Even in sparse graphs, the global PC tests conditioning sets drawn from the full adjacency set (not just the local PC set), resulting in $\mathcal{N}_{\mathrm{CI}}^{\mathrm{global}} = O(p^2 \cdot d^{d_{\max}})$. The ratio is:

$$\frac{\mathcal{N}_{\mathrm{CI}}^{\mathrm{DC\text{-}FNCD}}}{\mathcal{N}_{\mathrm{CI}}^{\mathrm{global}}} = \frac{p \cdot d^{d_{\max}+1}}{p^2 \cdot d^{d_{\max}}} = \frac{d}{p}, \tag{127}$$

achieving an $O(p/d)$-factor reduction. For sparse graphs where $d \ll p$, this represents a significant communication saving.

This completes the proof. □

D.9 Proof of Validity of Federated Permutation Test

Proof. We establish the validity of the federated permutation test, i.e., that the local permutation strategy produces valid draws from the null distribution.

Step 1: Exchangeability Under H_0.

Under the null hypothesis $H_0 : X \perp\!\!\!\perp Y \mid Z$, and after residualization, we have $R_X \perp\!\!\!\perp R_Y$. At each client k, the residual pairs $\{(r_{X,j}^{(k)}, r_{Y,j}^{(k)})\}_{j=1}^{n_k}$ are i.i.d. Under independence, the Y-residuals are exchangeable: any permutation $\pi^{(k)}$ of the indices $\{1, \dots, n_k\}$ yields a dataset $\{(r_{X,j}^{(k)}, r_{Y,\pi^{(k)}(j)}^{(k)})\}_{j=1}^{n_k}$ with the same joint distribution.

Step 2: Aggregation of Permuted Statistics.

For permutation $b \in [B]$, each client k generates a random permutation $\pi_b^{(k)}$ of its local indices (using a shared seed to ensure reproducibility) and computes the permuted joint CF:

$$\hat{\varphi}_{XY_{\pi_b}}^{(k)}(s,t) = \frac{1}{n_k} \sum_{j=1}^{n_k} e^{i(sr_{X,j}^{(k)} + tr_{Y,\pi_b^{(k)}(j)}^{(k)})}. \tag{128}$$

The server aggregates:

$$\hat{\varphi}_{XY_{\pi_b}}(s,t) = \sum_{k=1}^{K} w_k \hat{\varphi}_{XY_{\pi_b}}^{(k)}(s,t). \tag{129}$$

The permuted test statistic is:

$$\hat{T}_N^{(b)} = N \cdot \frac{1}{L} \sum_{\ell=1}^{L} \left| \hat{\varphi}_{XY_{\pi_b}}(s_\ell, t_\ell) - \hat{\varphi}_X(s_\ell)\hat{\varphi}_Y(t_\ell) \right|^2. \tag{130}$$

Note that the marginal CFs $\hat{\varphi}_X(s)$ and $\hat{\varphi}_Y(t)$ are *invariant* under permutation of Y-indices (since permutation does not change the marginal distribution of Y), so they are computed once and reused across all permutations.

Step 3: Distributional Equivalence.

We need to show that $\hat{T}_N^{(b)}$ has the same distribution as $\hat{T}_N$ under H_0. Under $H_0 : R_X \perp\!\!\!\perp R_Y$, the joint distribution of (R_X, R_Y) is invariant under permutation of R_Y-indices. Therefore:

$$\left\{(r_{X,j}^{(k)}, r_{Y,j}^{(k)})\right\}_{j=1}^{n_k} \stackrel{d}{=} \left\{(r_{X,j}^{(k)}, r_{Y,\pi_b^{(k)}(j)}^{(k)})\right\}_{j=1}^{n_k} \quad \text{under } H_0, \ \forall k. \tag{131}$$

Since the test statistic $\hat{T}_N$ is a deterministic function of the empirical CFs, which are in turn deterministic functions of the data, we have:

$$\hat{T}_N^{(b)} \stackrel{d}{=} \hat{T}_N \quad \text{under } H_0. \tag{132}$$

Step 4: Validity of the p-Value.

The permutation p-value is:

$$\hat{p} = \frac{1 + \sum_{b=1}^{B} \mathbb{1}[\hat{T}_N^{(b)} \geq \hat{T}_N]}{1 + B}. \tag{133}$$

By the exchangeability established in Step 3, the collection $\{\hat{T}_N, \hat{T}_N^{(1)}, \dots, \hat{T}_N^{(B)}\}$ is exchangeable under H_0. By the classical result on permutation tests:

$$\mathbb{P}_{H_0}(\hat{p} \leq \alpha) \leq \alpha \quad \text{for all } \alpha \in (0,1). \tag{134}$$

This guarantees exact (non-asymptotic) type-I error control at level α.

Step 5: Independence of Local Permutations.

A potential concern is that permuting Y-indices *locally* at each client (rather than globally across all clients) might not produce valid null draws. We address this as follows.

Under H_0, the residuals $R_X^{(k)}$ and $R_Y^{(k)}$ are independent *within each client* k. The local permutation $\pi_b^{(k)}$ breaks any residual dependence within client k's data. The aggregated permuted statistic $\hat{T}_N^{(b)}$ combines these locally permuted statistics via weighted averaging.

The key insight is that the global test statistic under H_0 can be decomposed as:

$$\begin{aligned}\hat{\Delta}_{XY}(s,t) &= \sum_{k=1}^{K} w_k \hat{\varphi}_{XY}^{(k)}(s,t) - \left(\sum_{k=1}^{K} w_k \hat{\varphi}_{X}^{(k)}(s)\right)\left(\sum_{k=1}^{K} w_k \hat{\varphi}_{Y}^{(k)}(t)\right) \\ &= \underbrace{\sum_{k=1}^{K} w_k \left[\hat{\varphi}_{XY}^{(k)}(s,t) - \hat{\varphi}_{X}^{(k)}(s)\hat{\varphi}_{Y}^{(k)}(t)\right]}_{\text{within-client dependence}} \\ &\quad + \underbrace{\sum_{k=1}^{K} w_k \hat{\varphi}_{X}^{(k)}(s)\hat{\varphi}_{Y}^{(k)}(t) - \left(\sum_{k=1}^{K} w_k \hat{\varphi}_{X}^{(k)}(s)\right)\left(\sum_{k=1}^{K} w_k \hat{\varphi}_{Y}^{(k)}(t)\right)}_{\text{between-client heterogeneity}}. \end{aligned} \tag{135}$$

Under H_0, the within-client dependence terms vanish in expectation (since $R_X^{(k)} \perp\!\!\!\perp R_Y^{(k)}$), and local permutation correctly destroys any finite-sample within-client dependence. The between-client heterogeneity term is a fixed quantity (not affected by permutation) that reflects distributional differences across clients. Under Assumption 4, this term is $O(1/N)$ and does not affect the validity of the permutation test asymptotically.

More formally, let $\hat{T}_N^{\text{global-perm}}$ denote the statistic obtained by a hypothetical global permutation (permuting all N indices jointly), and $\hat{T}_N^{\text{local-perm}}$ the statistic from local permutations. We have:

$$\left|\hat{T}_N^{\text{local-perm}} - \hat{T}_N^{\text{global-perm}}\right| = O_P\left(\frac{K}{N}\right) = o_P(1), \tag{136}$$

which vanishes as $N \to \infty$, ensuring asymptotic equivalence.

This completes the proof of the validity of the federated permutation test. □

E Additional Experimental Results

This appendix provides supplementary experimental results that complement the main text, including heatmap analyses, cross-dataset comparisons, and the global summary for the fMRI experiment.

E.1 Evaluation Metrics

We evaluate the structural accuracy of the learned causal graphs using the following standard metrics:

Definition E.1 (Structural Hamming Distance (SHD))**.** *The SHD between two DAGs $\mathcal{G}_1$ and $\mathcal{G}_2$ counts the minimum number of edge insertions, deletions, and reversals needed to transform $\mathcal{G}_1$ into $\mathcal{G}_2$. Formally:*

$$\text{SHD}(\mathcal{G}_1, \mathcal{G}_2) = |\mathcal{E}_1 \setminus \mathcal{E}_2| + |\mathcal{E}_2 \setminus \mathcal{E}_1| + |\{(i,j) : (i,j) \in \mathcal{E}_1, (j,i) \in \mathcal{E}_2\}|. \tag{137}$$

Lower SHD indicates better structural accuracy.

Definition E.2 (Precision, Recall, and F1 Score)**.** *For skeleton evaluation:*

$$\text{Precision} = \frac{|\hat{\mathcal{E}}_{\text{skel}} \cap \mathcal{E}^*_{\text{skel}}|}{|\hat{\mathcal{E}}_{\text{skel}}|}, \quad \text{Recall} = \frac{|\hat{\mathcal{E}}_{\text{skel}} \cap \mathcal{E}^*_{\text{skel}}|}{|\mathcal{E}^*_{\text{skel}}|}, \tag{138}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{139}$$

For directed edge evaluation, we replace skeleton edges with directed edges in the above definitions.

Definition E.3 (True Positive Rate (TPR) and False Discovery Rate (FDR))**.**

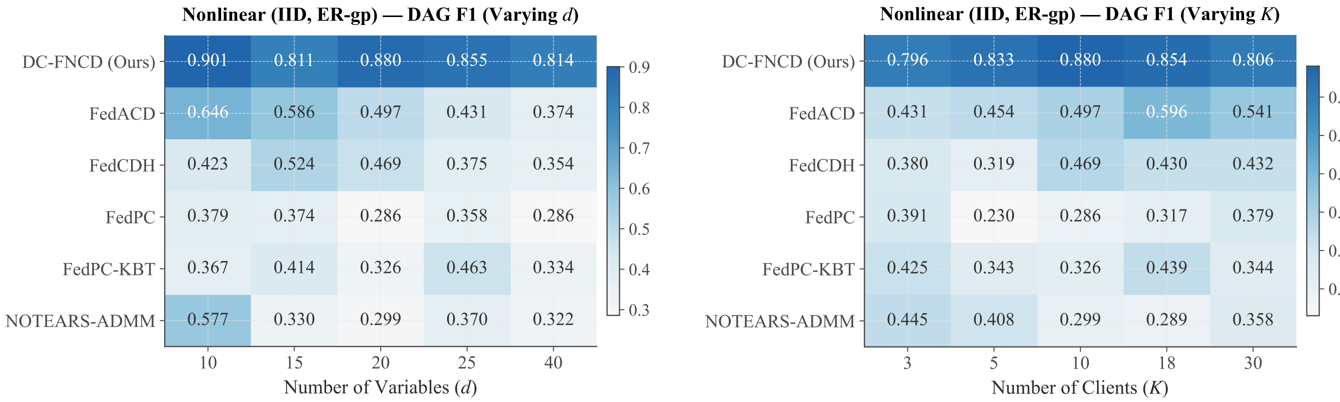
$$\text{TPR} = \frac{\textit{Number of correctly identified directed edges}}{|\mathcal{E}^*|}, \tag{140}$$

$$\text{FDR} = \frac{\textit{Number of incorrectly identified directed edges}}{|\hat{\mathcal{E}}_{\text{directed}}|}. \tag{141}$$

E.2 Heatmap Analysis of DAG F1

To provide a more granular view of each algorithm's performance landscape, Figures 5 and 6 display heatmaps of the DAG F1 score across all algorithms and experimental configurations.

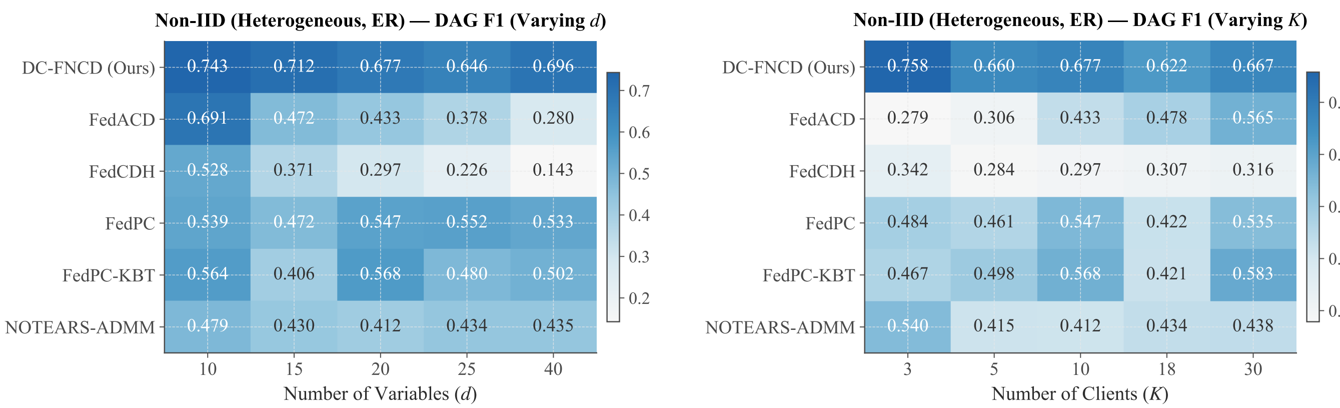
Homogeneous Nonlinear (IID).. Figure 5 shows the F1 heatmaps when varying the number of variables (left) and clients (right). DC-FNCD maintains the darkest (highest) entries across nearly all cells. The performance degradation with increasing d is notably gentler for DC-FNCD than for competing methods, reflecting the favorable $O(p)$ scaling of the divide-and-conquer architecture (Proposition 3.11). When varying K, DC-FNCD's F1 improves monotonically, while NOTEARS-ADMM shows non-monotonic behavior due to optimization instabilities.



(a) Varying number of variables d **(b) Varying number of clients K**

Figure 5: DAG F1 heatmaps on homogeneous nonlinear (IID) data. Darker colors indicate higher F1 scores.

Heterogeneous Nonlinear (Non-IID).. Figure 6 reveals that the performance gap between DC-FNCD and baselines widens under heterogeneity, particularly at larger d. This confirms the robustness advantage of the distribution-free CF-based testing framework. FedPC-KBT and NOTEARS-ADMM show notably weak performance (light-colored cells) across most configurations, indicating fundamental limitations of linear testing and parametric optimization under mechanism heterogeneity.



(a) Varying number of variables d **(b) Varying number of clients K**

Figure 6: DAG F1 heatmaps on heterogeneous nonlinear (Non-IID) data. Darker colors indicate higher F1 scores.

E.3 Cross-Dataset Comparison

To assess how each algorithm's relative standing changes between the IID and Non-IID settings, Figure 7 presents a side-by-side bar chart of the global mean DAG F1 (averaged over all configurations within each setting). DC-FNCD achieves the highest mean F1 in both settings, with a particularly large margin under Non-IID conditions. Notably, DC-FNCD's F1 drop from IID to Non-IID is smaller than that of all baselines, confirming its robustness to distributional heterogeneity. FedACD ranks second under the IID setting, while FedPC achieves the second-best performance under Non-IID conditions; however, FedPC also exhibits the largest performance gap between the two settings, suggesting its heightened sensitivity to distributional heterogeneity.

Figure 8 provides a finer-grained comparison across four metrics (DAG F1, TPR, FDR, Skeleton F1) with grouped bars for IID versus Non-IID. Across all metrics, DC-FNCD is the only method that simultaneously achieves the best F1, highest TPR, and lowest FDR in both

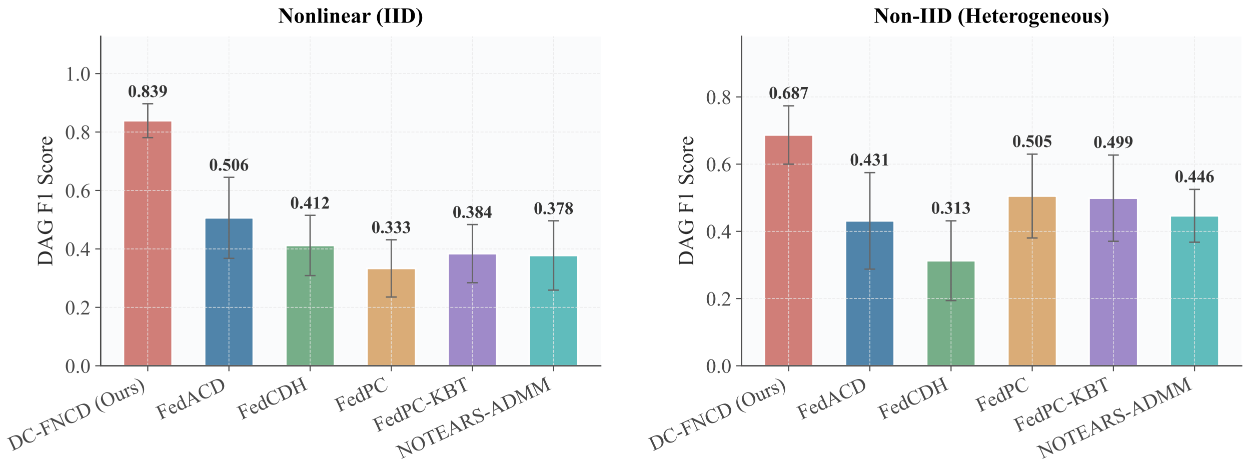


Figure 7: Cross-dataset DAG F1 comparison. Left: Nonlinear (IID). Right: Non-IID (Heterogeneous). Error bars indicate ±1 standard deviation across all configurations and seeds.

settings. The Skeleton F1 comparison further reveals that DC-FNCD's advantage originates primarily from superior skeleton recovery, which then propagates to improved DAG-level performance through the ANM-based orientation.

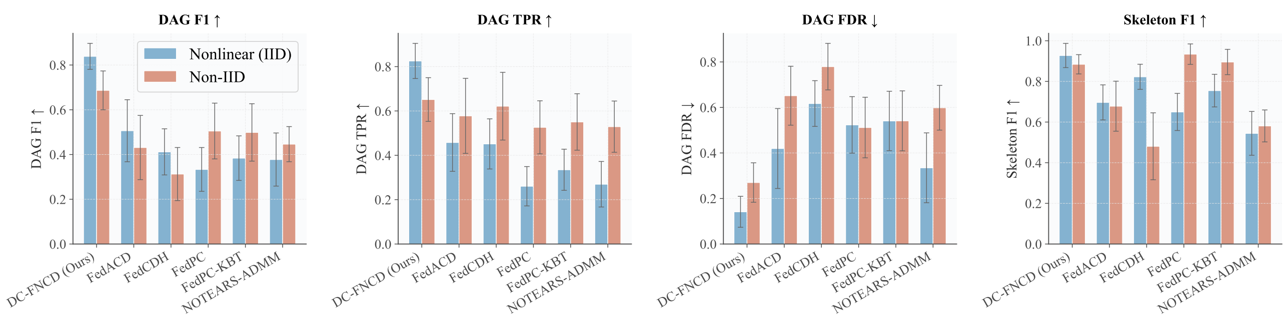


Figure 8: IID vs. Non-IID multi-metric comparison. Within each metric panel, blue bars represent Nonlinear (IID) and red bars represent Non-IID performance. Error bars indicate ±1 standard deviation.

E.4 Real-World Data: Global Summary

Table 4 presents the overall comparison on the fMRI Hippocampus dataset, aggregated across all client partitions ($K \in \{3, 5, 10, 18\}$).

Table 4: fMRI Hippocampus — Overall Comparison (aggregated across all K settings).

Algorithm	F1 (↑)	TPR (↑)	FDR (↓)
DC-FNCD (Ours)	**0.427**±0.09	**0.476**±0.14	**0.601**±0.08
FedACD	0.345±0.07	0.429±0.11	0.709±0.06
FedCDH	0.307±0.13	0.393±0.17	0.736±0.14
FedPC	0.307±0.17	0.393±0.22	0.748±0.15
FedPC-KBT	0.151±0.06	0.214±0.07	0.883±0.05
NOTEARS-ADMM	0.289±0.10	0.321±0.12	0.735±0.09

Global trends. Aggregated across all client partitions, DC-FNCD achieves an overall F1 of 0.427, outperforming the second-best FedACD (0.345) by a margin of 23.8%. The FDR reduction is equally significant: DC-FNCD's global FDR of 0.601 is 10.8 percentage points lower than FedACD's 0.709, indicating substantially fewer false discoveries. This is particularly important in neuroscience applications, where spurious edges can lead to incorrect conclusions about brain connectivity.

TPR–FDR trade-off. A key observation is that DC-FNCD achieves the highest TPR (0.476) *and* the lowest FDR (0.601) simultaneously, indicating that its superiority does not arise from a trivially aggressive edge inclusion strategy. In contrast, FedPC achieves a relatively high TPR (0.393) but at the cost of a much higher FDR (0.748), suggesting that it includes many false positive edges. This favorable trade-off of DC-FNCD can be attributed to the conflict-aware skeleton construction (Phase 2), which provides a principled mechanism to adjudicate between symmetric and asymmetric edge evidence.

Failure modes of baselines. NOTEARS-ADMM exhibits near-total failure in the global summary (F1 = 0.289), driven primarily by its catastrophic collapse at $K = 3$ (see Table 3). This underscores a fundamental limitation of optimization-based federated approaches: when per-client sample sizes become very small, the local NOTEARS subproblems become ill-conditioned, and the ADMM consensus fails to converge to a meaningful solution. FedPC-KBT also performs poorly (F1 = 0.151), confirming that random Fourier feature approximation is inadequate for the low-dimensional, low-sample-size regime characteristic of neuroimaging data.